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In re Patent Application of:	Reinaldo Javier Aguiar Marcano
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Title of Invention:	DISTRIBUTED SYSTEM AND METHOD FOR COVERT COMMUNICATION VIA SATELLITE-OBSERVABLE, GEO-SPATIALLY LOCATED VISUAL ENCODERS

REPLACEMENT ABSTRACT (37 CFR § 1.72(b))
(COMMENCING ON A SEPARATE SHEET)

ABSTRACT

A system and method for secure, untraceable communication and operational coordination of a distributed network of agents is disclosed. The system architecture utilizes a novel, two-tiered sharding protocol for dynamic task allocation to a plurality of mobile assets, such as vehicles. An 'object sharding' function first partitions assets into groups based on physical features, after which a time-variant 'function sharding' protocol dynamically assigns operational roles to said groups, creating a resilient and unpredictable operational pattern. Communication, including the transmission of status messages and encryption key seeds, is achieved without reliance on the conventional radio frequency (RF) spectrum. Instead, the system uses satellite-observable 'visual encoders'—arrangements of large physical objects of specific colors, shapes, and orientations—to transmit data. A highly secure, two-factor method for transmitting encryption key seeds is also disclosed, wherein a first seed component is transmitted via a satellite-observable visual encoder at a first geographical location, and a second component is transmitted via a local, non-RF channel at a second location. The first location is mathematically derived from the second by applying a predetermined coordinate transformation, such as negating the longitude, thus obviating the need for a central database of keying locations and providing tiered security.